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ODD END SEMESTER EXAMINATION DEC 2024

Name of the Course: B.Tech Civil Engineering

Semester: III

Name of the Paper: Geomatic Engineering

Paper Code: TCE 302

Time: 3 Hours

Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1	(20marks)												
(a)	(i) Describe the principle of surveying? Discuss in detail. (ii) Design a vernier for a theodolite circle divided into degrees and half degrees to read up to 30''.	CO 1											
(b)	(i) What are different kinds of error? Discuss in details. (ii) A surveyor measured the distance between two points on the plan drawn to a scale of 1cm=40m and the result was 468m. later however, he discovered that he used a scale of 1 cm=20m. find the true distance between the points.												
(c)	The plan of an area has shrunk such that a line originally 10 cm now measures 9.8 cm. The original scale of the plan was 1 cm = 10 m. Determine the following: (i) Shrinkage factor (ii) Shrunk scale (iii) Correct distance corresponding to a measured distance of 90m (iv) Correct area corresponding to a measured area of 14000 m ²												
Q2	(20marks)												
(a)	Define the terms : (i) True bearing (ii) Magnetic bearing (iii) Local attraction (iv) Back bearing (v) Magnetic declination	CO 2											
(b)	(i) Convert the following quadrantal bearing to whole circle bearings: (a) N 12 ⁰ 24' E (b) S 31 ⁰ 36' E (c) S 68 ⁰ 6' W (d) N 5 ⁰ 42' W (ii) Find the magnetic declination at a place if the magnetic bearing of the sun at noon is (a) 184 ⁰ (b) 350 ⁰ 20'												
(c)	(i) The following bearings were observed with a compass. Calculate the interior angles. <table><tr><td>LINE</td><td>FORE BEARING</td></tr><tr><td>AB</td><td>60⁰30'</td></tr><tr><td>BC</td><td>122⁰0'</td></tr><tr><td>CD</td><td>46⁰0'</td></tr><tr><td>DE</td><td>205⁰30'</td></tr><tr><td>EA</td><td>300⁰0'</td></tr></table>		LINE	FORE BEARING	AB	60 ⁰ 30'	BC	122 ⁰ 0'	CD	46 ⁰ 0'	DE	205 ⁰ 30'	EA
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Q3	(20marks)												
(a)	Discuss in detail the special methods of Spirit Levelling.	CO 3											
(b)	What is contour interval? Explain the methods of locating contours.												
(c)	The following staff reading were observed successively with a level, the instrument having been moved after 3 rd , 6 th and 8 th readings : 2.228, 1.606, 0.988, 2.090, 2.864,												

	1.262, 0.602, 1.982, 1.044, 2.684 metres. Enter this reading in a level book and calculate the R.L of each point if the first reading was taken with a staff held on a bench mark of 432.384 m. Use Rise and Fall method. Also provide arithmetic check.	
Q4	(20marks)	CO 4
(a)	What are the working operations of Plane table Survey? Discuss each in detail.	
(b)	Distinguish between the following methods of Plane tabling: (i) Radiation (ii) Intersection	
(c)	What are the advantages and disadvantages of Plane tabling?	
Q5	(20marks)	CO 5
(a)	Distinguish clearly between cumulative and compensating error.	
(b)	Calculate the area of rectangle if its sides are given as : A(10 m $\pm$.02 m) B(15m $\pm$.03 m)	
(c)	In carrying a line of levels across a river, the following eight readings were taken with a level under identical conditions : 2.322, 2.346, 2.352, 2.306, 2.312, 2.300, 2.306, 2.326 metres. Calculate : (i) The probable error of single observation (ii) Probable error of the mean.	